

GentleWoman Headless Commerce

Storefront (Frontend) × Odoo 19 (Backend) — Integrated Architecture

Business Presentation · Product Owner Value-Added Services · v1.0 · Juni 2026

Executive Summary

Headless commerce: Storefront Next.js mengontrol pengalaman brand; Odoo 19 menjadi single source of truth.

- Brand-controlled editorial UX (Next.js 15) — Odoo tetap pemilik catalog, stock, price, order, customer
- Diferensiasi pengalaman fashion: 3D Product Detail Page + AI Personal Shopper
- Integrasi aman 3-lapis lewat BFF: Public (CORS) · Customer (JWT) · Admin (HMAC)
- Delivery cepat: maksimal reuse modul platform + AI-agentic development (Claude Code)

3D

Product Viewer

AI

Personal Shopper

3-Layer

Secure Integration

~107

Mandays

Solution Overview — Frontend × Backend

Storefront — Frontend (Next.js 15)

- Presentasi & editorial design (Tailwind + Framer Motion)
- 3D Product Detail Page (three.js / React-Three-Fiber)
- AI Personal Shopper widget
- Multilanguage ID/EN (URL-prefixed)
- BFF route handlers (server-side, inject auth + tenant)
- Session: sealed HttpOnly cookies (AES-256-GCM)

Odoo 19 — Backend (System of Record)

- Catalog, varian/ukuran, tags, material
- Pricelist & promo (strike-through, diskon %)
- Stock & in-store availability (per warehouse)
- Cart & checkout = sale.order (pajak native)
- Customer, address book, guest checkout
- Affiliate, editorial content, fulfilment & tax

Prinsip headless: Frontend mengontrol presentasi; Odoo mengontrol data & proses bisnis (single source of truth).

Feature Catalog — Storefront Capabilities

Catalog & PDP / Katalog

PLP filter (kategori, tag, range harga), search, sort, pagination · PDP: galeri, varian/ukuran, material, ref/SKU, badge 'New' otomatis

AI Personal Shopper

Chat stylist · intent → retrieve katalog Odoo → synthesize · anti-halusinasi (produk real) · eskalasi ke manusia (WhatsApp)

Store Locator + Click & Collect

Toko = stock.warehouse published (geo, jam buka) · stok per toko · ambil di toko (pickup XOR kirim)

Affiliate Program

Link & klik tracking · atribusi ke order · komisi & payout · dashboard self-serve afiliasi

3D Product Viewer

three.js + React-Three-Fiber · gambar produk pada mesh kain · drag-rotate · toggle 2D■3D · CSP-safe tanpa fetch eksternal

Cart · Checkout · Wishlist

Cart = draft sale.order (pricelist/pajak native) · checkout guest & member · alamat kirim/tagih · wishlist + move-to-cart

Promo & Multilanguage

Strike-through price dari pricelist + diskon % · ID/EN URL-prefixed, field translatable Odoo, OG tag terlokalisasi

Editorial CMS + Newsletter

Blok konten (hero/editorial/lookbook) dikelola di Odoo · newsletter subscriber · consent UU PDP

3D Product Detail Page

Bagaimana 3D PDP Bekerja

- three.js + React-Three-Fiber + drei
- Gambar produk dipetakan ke mesh kain (sine-drape)
- Drag-rotate via PresentationControls
- Contact shadow + HDRI environment (self-hosted)
- Lazy-load (SSR off), toggle 2D■3D di PDP

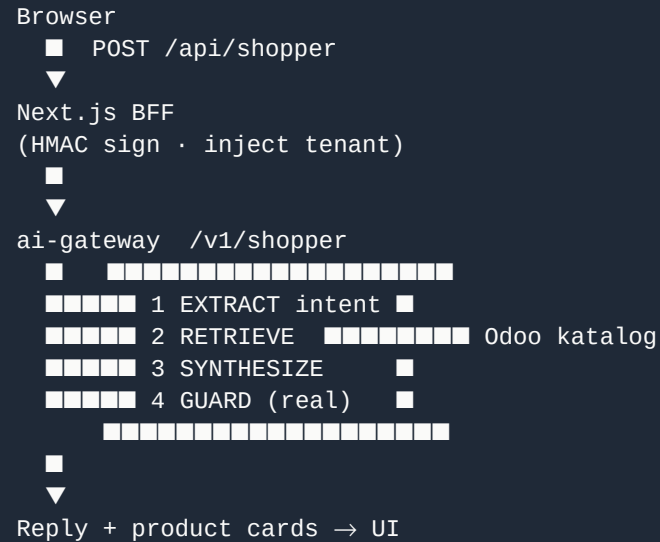
Keamanan & Performa

- CSP-safe: gambar lewat proxy /api/img (tanpa fetch eksternal)
- Bump map prosedural agar tekstur tidak 'plastik'
- Aset HDRI self-hosted (tanpa CDN pihak ketiga)
- Render hanya saat viewer dibuka
- Fallback galeri 2D bila perangkat terbatas

Fashion saat ini memakai image-on-cloth; hard goods (tas/sepatu) dapat memakai model 3D nyata pada fase berikutnya.

AI Personal Shopper — Retrieval-Controlled

Bukan free-form: jawaban & kartu produk selalu berasal dari katalog Odoo yang nyata (anti-halusinasi).

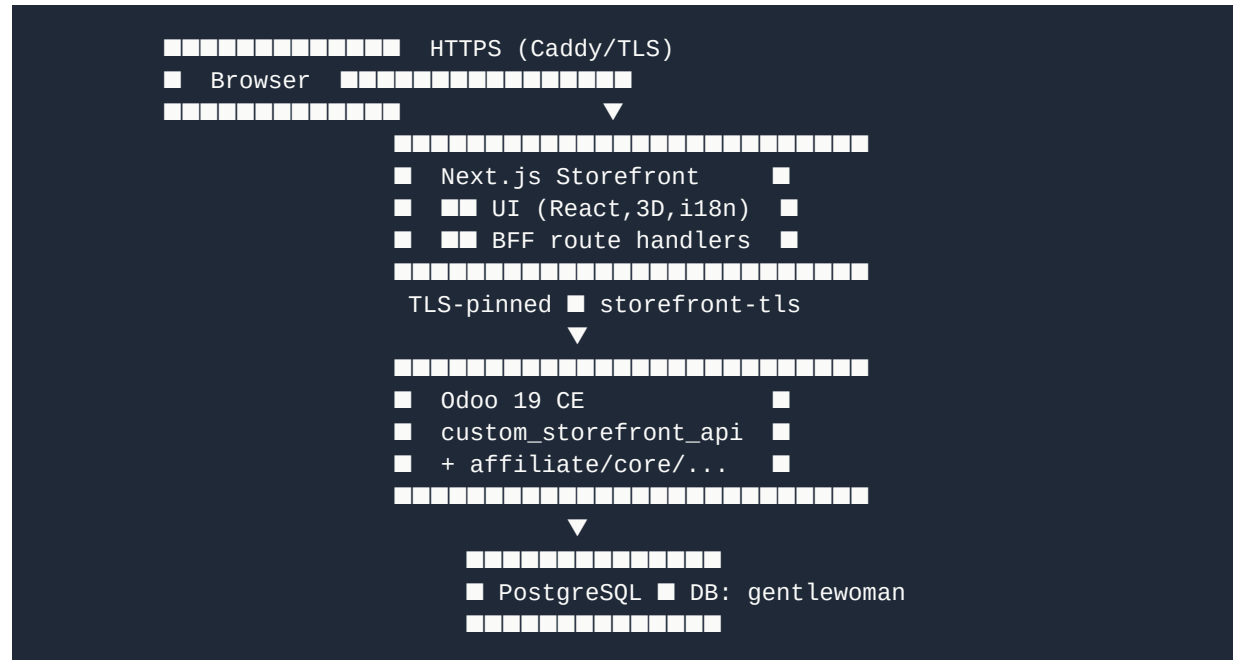


Key Decisions

- **EXTRACT** — parse intent (occasion, budget, warna, ukuran), JSON-mode
- **CLARIFY** — jika terlalu umum, tanya balik (tidak menebak)
- **RETRIEVE** — ambil produk dari Odoo /storefront/api/products sesuai filter
- **SYNTHESIZE** — susun balasan natural dari item nyata
- **GUARD** — kartu produk hanya dari payload katalog (anti-halusinasi)
- **ESCALATE** — komplain/retur diarahkan ke manusia (WhatsApp)
- **Provider**: ai-gateway (Claude / Ollama lokal) · HMAC-signed · prompt caching

Integration Architecture — Storefront × Odoo

Browser tidak pernah bicara langsung ke Odoo — semua lewat BFF server-side (aman, tenant-scoped).



Key Decisions

- Tenant resolution: BFF kirim header X-Odoo-Database: gentlewoman (dbfilter ^%d\$)
- Single source of truth: Odoo pemilik catalog/stock/price/order/customer/promo/content
- BFF = security boundary: inject auth & tenant header server-side; secret tak pernah ke browser
- Internal hop BFF→Odoo lewat TLS sidecar (cert-pinned) — token/PII terenkripsi
- Reuse native: cart = website_sale · sale.order · shipping = custom_ecommerce · payment = custom_payment_id

Integration — 3 Auth Layers

Tiga lapis otentikasi sesuai sensitivitas data.

Layer	Mechanism	Scope / Data	Contoh Endpoint
Public	auth=public + CORS, read-only	Katalog, PDP, tags, stores, content, shipping quote	GET /storefront/api/products · /categories · /stores
Customer	JWT (HS256) bearer, sealed HttpOnly cookie	Per-customer: cart, order, address, wishlist (partner-scoped)	POST /auth/login · GET/POST /cart · POST /checkout
Admin	HMAC-SHA256 (X-Signature/X-Timestamp), server-to-server	Webhook BFF↔Odoo (health, product sync, order status)	POST /admin/health · /admin/sync/products

Note: JWT: partner di-resolve dari email claim → request.jwt_partner_id; body partner tak pernah dipercaya (anti-IDOR). HMAC replay window 5 menit.

Data Flow — Read & Write Path

Read Path — Catalog / Price / Stock

- Product di-serialize: name, price (pricelist), compare_at, discount_pct
- Tags, material, is_new, in_stock per varian
- Pricelist di-resolve sekali per request (mata uang IDR)
- Stok: qty_available; in-store per warehouse (location-scoped)
- Gambar lewat proxy /api/img (anti-SSRF); lang=id|en field translatable

Write Path — Cart → Order

- Add to cart → draft sale.order (_cart_add native: pricelist + pajak)
- Pilih kurir (custom_ecommerce) atau pickup toko (XOR)
- Checkout → sale.order.action_confirm → order terkonfirmasi
- Bayar → payment.transaction (custom_payment_id; Eraspace adapter stub)
- Token JWT AES-256-GCM sealed, refresh transparan; affiliate attribution di order

Odoo Backend Responsibilities

Module / Service	Tanggung Jawab
custom_storefront_api	API publik/JWT/HMAC, cart/checkout, content, store locator, PII-at-rest, guest checkout
custom_affiliate	Master afiliasi, link/klik/konversi/payout, atribusi order, cron harian
custom_core	Platform: HMAC secure-endpoint, Fernet field encryption helpers
custom_ecommerce	Kurir Indonesia (JNE/JNT/SiCepat/AnterAja...) + rate shipment
custom_payment_id	Payment gateway adapters (Midtrans/Xendit; Eraspaces stub)
custom_pdp_* (6 modul)	Klasifikasi data, consent, audit, masking (UU PDP)
ai-gateway (sidecar)	Abstraksi multi-provider (Claude/Ollama), shopper retrieval, HMAC + rate limit

Security & Privacy

Application & Session

- Strict Content-Security-Policy + security headers
- Token HttpOnly + AES-256-GCM sealed; refresh transparan
- Anti-IDOR: ownership check alamat (kirim/tagih)
- Rate-limit + body cap pada endpoint sensitif
- HTML sanitize (DOMPurify) · no raw SQL (ORM only)

Transport, Data & Compliance

- HTTPS (Caddy) + internal TLS sidecar cert-pinned
- PII Fernet at-rest (phone/street/zip)
- Anti-SSRF image proxy (/api/img dikunci ke /web/image)
- UU PDP: consent capture + cookie banner
- Secrets externalized + rotation · DB-per-tenant isolation

Delivery Accelerators — Reuse + AI-Agentic

Platform Reuse — Sudah Tersedia

- custom_storefront_api — scaffolding API headless (3-lapis auth)
- custom_core — HMAC & Fernet encryption helpers
- custom_pdp_* — consent, audit, masking (UU PDP)
- custom_ecommerce — kurir & rate shipment Indonesia
- custom_payment_id — adapter payment gateway
- website_sale native — cart, pricelist, pajak
- ai-gateway — multi-provider (Claude/Ollama) siap pakai

AI-Agentic Delivery (Claude Code)

- Generate controller, model, boilerplate API
- Scaffolding komponen Next.js & unit test
- Drafting dokumentasi & spec
- Refactor & review berbantuan AI
- Lebih sedikit kode tulis-tangan → mandays Developer turun
- PMO cukup footprint tata kelola (bukan full-time)

Estimasi mandays mengasumsikan reuse maksimal + AI-agentic development. Build tradisional setara ~200–300 MD.

Implementation Mandays by Role

Estimasi effort (bukan biaya) untuk GentleWoman Storefront + integrasi Odoo. Kolom = PMO · IT BA · IT Developer · QA.

Phase	PMO	IT BA	IT Dev	QA
1. Discovery & Requirements (BRD/FSD)	2	6	1	1
2. Solution & Technical Design (Blueprint/TSD)	2	3	4	1
3. Odoo Backend API (extend custom_storefront_api)	1	2	11	3
4. Storefront Build (Next.js, AI-accelerated)	1	2	14	4
5. 3D PDP (three.js / R3F)	1	0	5	1
6. AI Personal Shopper (ai-gateway glue)	1	1	6	2
7. Integration & Security Hardening	1	1	6	3
8. UAT, Bug-fix & Stabilization	1	3	4	5
9. Deploy, Go-live & Hypercare	2	1	3	2
TOTAL (mandays)	12	19	54	22

Note: Total ~107 mandays. IT Developer mencakup backend Odoo + frontend Next.js + 3D + AI-gateway glue + DevOps (paralel ~2 developer). Re-baseline saat BRD freeze.

Project Timeline (~10 Minggu)

Minggu 1–2	Minggu 3–4	Minggu 5–6	Minggu 7–8	Minggu 9–10
● Discovery & Requirements (BRD/FSD) ● Solution & Technical Design (mulai) ● Setup environment & akses	● Technical Design selesai (Blueprint/TSD) ● Odoo Backend API (extend storefront_api) ● Storefront Build mulai (PLP/PDP)	● Storefront Build (cart, checkout, i18n) ● 3D PDP ● AI Personal Shopper (ai-gateway glue)	● AI Shopper selesai ● Integration & Security Hardening ● UAT mulai	● UAT & Bug-fix ● Deploy, Go-live & Hypercare ● Handover & dokumentasi

Delivery Scope & Status

Status fitur saat ini (Phase-1 live pilot di tenant gentlewoman).

Status	Fitur
Delivered	Catalog/PLP/PDP · 3D PDP · AI Shopper (MVP) · Cart/Checkout/Wishlist · Store Locator + Click&Collect; · Affiliate · Multilanguage ID/EN · Promo · Security & PDP consent
Pending	CMS lanjutan · Share-to-social lengkap · AI Admin (auto-deskripsi/tagging) · Avatar / Virtual Try-On (Fase B/C)
Blocked	Payment gateway Eraspaces — menunggu dokumentasi API vendor (checkout siap di belakang PaymentProvider interface + feature flag)

Risks & Mitigations

Risk	Mitigation
Payment Eraspaces tertunda (API vendor)	Checkout sampai pembuatan order di belakang interface + feature flag; aktif begitu API tersedia
Biaya AI membengkak	Prompt caching + rate limit per tenant + fallback Ollama lokal
Kebocoran data antar-tenant	DB-per-tenant + HMAC + tenant allow-list + header X-Odoo-Database
Scope creep menggeser mandays	BRD freeze sebelum build · change request via approval engine · re-baseline
Rate-limiter in-memory (single instance)	Pindah ke Redis saat scale multi-instance

Document Control & Pengesahan

Versioning & pengesahan dokumen presentasi.

Item	Detail
Document	GentleWoman Headless Commerce — Business Presentation
Document ID / Version	GW-PRES-001 · v1.0
Status / Classification	Final for Approval · Confidential — Internal
Date	Juni 2026
Prepared by	Product Owner — Value-Added Services (Erajaya)
Reviewed by	IT Business Analyst / Technical Lead — _____
Approved by	Head of Value-Added Services / IT — _____

Note: Riwayat revisi: v1.0 — rilis awal (Juni 2026).

Terima Kasih

Terima Kasih

GentleWoman Headless Commerce — Storefront × Odoo · Diskusi & Q&A;

Product Owner — Value-Added Services · Erajaya · v1.0 · Juni 2026